

Salesforce CLI

The modern `sf` command set for orgs, metadata, data, and DevOps workflows.

• Install & Update

```
npm install -g @salesforce/cli
sf update
sf version --verbose
sf --help
```

- Command shape: `sf <topic> <command> --flags`.
- Add `--json` to any command for scriptable output.

• Authorize Orgs

```
sf org login web --alias dev
sf org login web --alias prod \
  --instance-url
https://login.salesforce.com
sf org list
sf config set target-org dev
```

- `--set-default` (`-d`) makes it the default org.
- JWT flow for CI: `sf org login jwt`.

• Create Scratch Orgs

```
sf org create scratch \
  --definition-file config/project-scratch-
def.json \
  --alias scratch --set-default --duration-
days 7

sf org open --target-org scratch
sf org delete scratch --target-org scratch
```

• Deploy & Retrieve

```
sf project deploy start --source-dir force-
app
sf project deploy start --manifest
package.xml
sf project retrieve start --metadata
ApexClass

sf project deploy validate --source-dir
force-app
sf project deploy quick --job-id <id>
```

- `--dry-run` checks without saving.
- `--test-level RunLocalTests` for prod.

• Source Tracking

```
sf project deploy preview
sf project retrieve start      # pull org
changes
sf project deploy start      # push
local changes
```

- Scratch orgs & sandboxes track source diffs automatically.

• Run Apex & Tests

```
sf apex run --file script.apex
sf apex run test --code-coverage \
  --result-format human --wait 10
sf apex tail log --color
sf apex get log --number 1
```

• Query & Manipulate Data

```
sf data query --query "SELECT Id, Name FROM
Account"
sf data create record --subject Account \
  --values "Name='Acme'"
sf data import bulk --file accts.csv --
subject Account
sf data upsert bulk --file a.csv --subject
Account \
  --external-id Ext_Id__c
sf data export bulk --query "..." --output-
file out.csv
sf data export tree --query "SELECT Id FROM
Account"
```

• Packages

```
sf package create --name MyPkg --package-
type Unlocked
sf package version create --package MyPkg \
  --installation-key-bypass --wait 20
sf package install --package 04t... --wait
10
```

• Project & Org Scaffolding

```
sf project generate --name my-app
sf org generate password --target-org
scratch
sf subject describe --subject Account
sf org display --verbose
```

• Handy Flags

FLAG	PURPOSE
<code>--target-org</code> <code>-o</code>	choose the org
<code>--json</code>	machine-readable output
<code>--wait</code> <code>-w</code>	minutes to wait
<code>--loglevel</code>	debug verbosity

Set defaults once with `sf config set target-org` and `target-dev-hub` to skip `-o` on every call.

• Agentforce & AI

```
sf agent generate agent-spec
sf agent generate authoring-bundle --spec
spec.yaml
sf agent preview --api-name My_Agent
sf agent test run --api-name My_Suite
```

- The authoring-bundle flow builds an Agent Script `.agent`; `sf agent create` is the legacy (non-Agent-Script) path.
- `sf` unifies dev, data, DevOps, and Agentforce workflows in one CLI.